

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459976.9413 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.029
Transit depth (Rp/Rs)^2	2.21 +/- 0.85 [%]
Orbital Inclination (inc)	83.95 +/- 1.98
Airmass coefficient 1 (a1)	1.276 +/- 0.011
Airmass coefficient 2 (a2)	-0.182 +/- 0.068
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	None
Optimal Aperture	4.05
Optimal Annulus	10.71
Transit Duration (day)	0.0419 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.029 vs 0.1594 (deviation 0.36σ)
Duration fit vs published	0.0419 d vs 0.0512 d (deviation 2.83σ)
Mid-time O–C	-5.2 min
Depth SNR	5.1
Residual scatter	0.82 %

Light curve

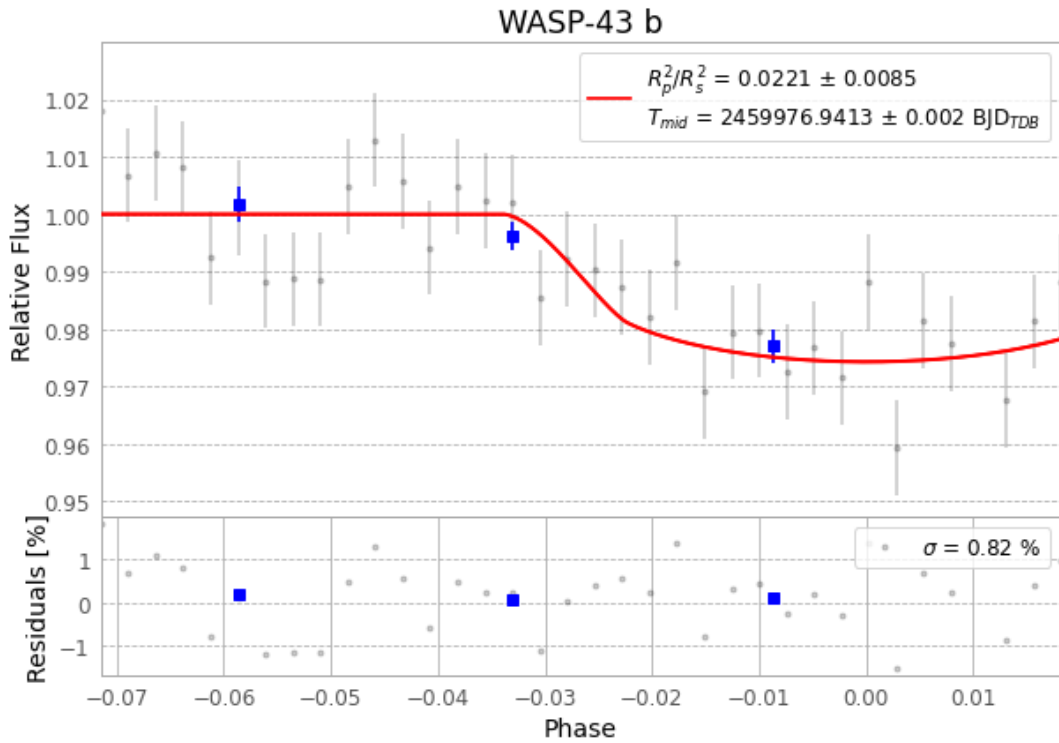


Figure 1 — FinalLightCurve_WASP-43 b_2023-02-01.png (SHA-256 b45dc9583151e219...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [448, 214], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e0518a7846bcdcf6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

36 FITS frames (24 MB), WASP-43230201090315.FITS → WASP-43230201104815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230201.log (37bf87c8dfc7...), FinalLightCurve_WASP-43 b_2023-02-01.png (b45dc9583151...), FinalLightCurve_WASP-43 b_2023-02-01.csv (0290a5640a3b...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 18:00Z.